

Apoorva Tyagi

Software Developer

apoorvatyagi61@gmail.com | +1-806-500-4556 | Hillsboro, OR

<https://www.linkedin.com/in/apoorvatyagi61/>

SUMMARY

Full-stack software engineer with 4+ years of experience building scalable, accessible web applications and AI-driven systems, with particular depth in Vue.js frontends, Node.js/Express APIs, and PostgreSQL-backed services. Proven track record delivering performant, secure, and real-time data solutions across HealthTech, logistics, AI/ML, and enterprise SaaS. Experienced building dashboards, score/reporting interfaces, and data visualizations for both technical and non-technical stakeholders

PROFESSIONAL EXPERIENCE

Intel Corporation

Software Developer II | Hillsboro, OR

May 2025 – June 2026

- Led full-stack development of BI dashboards for Intel's Client Computing Admin Portal, architecting real-time analytics across compute instances, reservations, and resource requests, which reduced allocation time by **40%** and improved capacity planning for DevCloud.
- Re-architected Client Instance Attributes DB (Vue.js, FastAPI, MongoDB), reducing per-machine update time by **80%** (10 min -> 2 min), which enabled more system entries per hour.
- Developed AI Workbench in Vue.js for real-time inference monitoring across GPU/CPU/NPU platforms, cutting benchmark analysis cycles by **50%** and accelerating hardware selection decisions.
- Extended the Engineering Kits UI for **Software DevOps Center (SDC)** with full ingredient lifecycle management and base OS version configuration, streamlining kit setup workflows for firmware teams across multiple platform families.
- Eliminated unintended resource duplication during ingredient creation by integrating one BKC node server with APIGEE, preventing unintended IPX cloning from tip-of-line.
- Resolved critical production bugs across kits and ingredients pipelines, including manifest SHA mismatches, XML parsing errors, upload failures, and search, which improved platform stability.

Alice Consulting LLC

Software Engineer | McKinney, TX

August 2024 – April 2025

- Designed and developed a secure data portal for a HealthTech client using **React.js + TypeScript**, integrating with EMR **RESTful APIs** (FHIR/HL7) to enable clinicians/researchers access to patient records with HIPAA-compliant authentication and role-based permissions.
- Created an interactive data visualization module using D3.js and React to present healthcare and operational metrics, enabling stakeholders to identify trends and make evidence-based decisions faster.

Quext

Lead Software Developer – Conversational AI | Lubbock, TX

Nov 2022 – June 2024

- Led the integration of Amazon Bedrock and Claude LLM into Quext's core products like Digital Human and Knowledge Library and collaborated with Amazon's Gen AI team to enhance user personalization and conversational depth.
- Implemented RAG (Retrieval-Augmented Generation) architecture (LangChain pipelines, Pinecone vector DB, retrieval tuning) for context-aware LLMs, boosting exact-match accuracy by +24% and cutting p95 latency by -31%.
- Optimized STT/TTS path on AWS (Transcribe, Polly) using chunked audio, connection reuse, and SSML tuning, improving agent responsiveness for real-time speech-to-text and text-to-speech conversion by 7 seconds (10 seconds → 3 seconds).
- Leveraged Google Cloud Platform's Dialogflow to design and tune conversational intents and NLU flows for Quext's Digital Human product.

ITHC Department - Texas Tech University

Software Developer - Graduate Assistant | Lubbock, TX

June 2021 – May 2022

- Implemented end-to-end ServiceNow ITSM enhancements for askIT (Service Portal, Workflows, Catalog, Notifications, and Surveys) with AngularJS/AJAX client scripts and business rules, resulting in a +25% resolution rate and a -30% open tickets.
- Accelerated reporting for the room reservation service by optimizing MySQL queries, improving utilization report generation by 45% for campus facilities scheduling.

FASY Solutions

Associate Developer | Mumbai, India

Jan 2020 – Dec 2020

- Built a mobile-responsive delivery management UI for a client in React with a WebGL/three.js live map and GraphQL subscriptions for real-time courier tracking and automated dispatch, which grew monthly active users by ~600.
- Engineered a scalable Node.js + MongoDB backend with GraphQL and utilized geospatial (2dsphere) queries for routing/ETAs to cut average delivery time by ~20 min.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, Java, SQL (PostgreSQL, MySQL), NoSQL (MongoDB)
Web Development	Vue.js 3 (Composition API), React.js, AngularJS, Node.js, Express.js, FastAPI, GraphQL, RESTful APIs, D3.js, three.js, HTML, SCSS
Generative AI	LLMs, AWS Bedrock, LangChain, RAG, Pinecone (Vector DB), Google Dialogflow, Prompt Engineering & Chaining
DevOps and Add-On	GCP, AWS (Lambda, S3, Bedrock, Transcribe, Polly), Docker, Kubernetes, Jenkins, ArgoCD, CI/CD pipelines, Git, Postman

EDUCATION

Westcliff University, San Francisco, CA
Professional MBA - Applied AI in Business
Texas Tech University, Lubbock, TX
Master of Science, Computer Science (**GPA: 3.7/4.0**)
Lovely Professional University, Punjab, India
Bachelor of Technology, Computer Science with Minor in Data Science (**CGPA: 9.5/10**)

PROJECTS

- Medical Assistant System** | TF-IDF, Cosine Similarity, NLP, Speech Synthesis
- Developed a medical voice assistant leveraging TF-IDF and cosine similarity to map symptoms to conditions with precision.
 - Integrated NLP pipeline with text-to-speech for interactive patient engagement.
 - Achieved 95% diagnostic accuracy, demonstrating the robustness of lightweight models compared to heavier ML systems.
- Dreamt It, Seen It** | OpenAI GPT-3.5, GraphQL, LangChain, Instagram Graph API
- Built a web platform to inspire fashion creators by generating text-based design concepts using GPT-3.5. Orchestrated multi-step prompt flows via LangChain for idea refinement.
 - Connected Instagram Graph API to retrieve live trend data, enabling real-time alignment between imagination and current fashion culture.
- Air Pollution Health Analysis** | Angular, D3.js, TensorFlow
- Implemented an interactive visualization system with Angular + D3.js to track air quality data from 190+ stations.
 - Built predictive models in TensorFlow to forecast respiratory disease incidence based on pollution factors.
 - Conducted hyperparameter tuning and validation, achieving 86% forecast accuracy.
- GPU-Accelerated Microservices** | Python, Docker, Kubernetes
- Designed and deployed containerized microservices optimized for GPU-based computation across multi-region Kubernetes clusters. Achieved scalable high-throughput graphics and data processing, ensuring reliability and low-latency availability.
 - Enabled cost-effective workload distribution by leveraging orchestration strategies across heterogeneous hardware.

PUBLICATIONS

- Cloud Analytics: An Outline of Tools and Practices — *Springer*, Apr 2022 [\[Link\]](#)
- Design of Cloud-based Interoperable Electronic Health Record with Advanced Security — *IEEE*, Oct 2021 [\[Link\]](#)
- Analyzing Air Pollution Mitigation Methodology: Odd-Even Campaign — *JETIR*, Dec 2019 [\[Link\]](#)